

Asmita Singh

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SUMMARY

Data Engineer and Software Developer with expertise in designing scalable ETL pipelines, data models, and real-time analytics on AWS. Skilled in Python, SQL, Redshift, Spark, and Java/Spring Boot for backend and data-intensive applications. Hands-on with BI workflows (Tableau, Quicksight) and cloud-native architectures (Docker, Kubernetes, RabbitMQ). Adept at translating large, complex datasets into actionable insights while delivering secure, scalable, and compliant solutions.

TECHNICAL SKILLS

Languages and Database: Python, Java, C | SQL, PL/SQL, MySQL, PostgreSQL, Oracle, MongoDB

Frameworks & Libraries: Spring Boot, Hibernate | React, Angular | JSP/Servlets | Oracle APEX | XML/JSON data processing

Data Engineering & Analytics: Spark, Hadoop, Redshift, BI & Visualization, Airflow | Tableau, Quicksight, Data Modeling, ETL Pipelines

Cloud & DevOps: AWS, Azure | Docker, Kubernetes | OAuth2 | Jenkins, CI/CD pipelines | RabbitMQ | Git

Testing & Performance: JUnit, Mockito, Selenium, | Query performance tuning | Load testing

Data & AI/ML Tools (Academic/Projects): PyTorch, TensorFlow | NLP techniques | Mistral-7B

PROFESSIONAL EXPERIENCE

Software Developer, Siemens Technology and Services Pvt Ltd 07/2019 – 07/2023 | Bangalore, India

- Architected and developed microservices with Java/Spring Boot, reducing API latency by 30% (500ms→350ms) and enabling real-time processing on high-traffic platforms.
- Designed and optimized scalable ETL pipelines into Amazon Redshift, ensuring SLA compliance, data quality, and high-performance analytics supporting business stakeholders.
- Built internal analytics applications in Oracle APEX, boosting procurement workflow visibility and decision-making.
- Designed and executed scalable data ingestion pipelines and automated reporting workflows on multi-million row datasets using Power Automate and Tableau, improving data accessibility and real-time analytics.
- Developed and optimized ETL pipelines into Amazon Redshift, ensuring data quality, SLA compliance, and efficient analytics for business stakeholders.
- Drove end-to-end SDLC in global Agile teams: requirements gathering, architecture/design, CI/CD implementation, and rigorous code reviews for scalable deliverables.
- Ensured data governance and compliance by defining SLAs, validating pipeline quality, and aligning architecture with enterprise security standards.

Programmer Analyst, Cognizant Technology Solutions 06/2016 – 06/2019 | Chennai, India

- Maintained and enhanced legacy enterprise web applications using Java 7, JSP, and Servlets, while modernizing front-end components with AngularJS and jQuery to improve performance and user experience.
- Developed scalable RESTful APIs and optimized relational database schemas (PostgreSQL, Oracle, MySQL), improving throughput by 30%.
- Developed an AI-driven claims investigation module with OAuth2 authentication, significantly improving claim loading efficiency.
- Improved PostgreSQL transaction throughput and backend service reliability through query profiling and advanced tuning.
- Authored HLD and LLD documentation aligning system design with data governance, privacy, and security requirements.

EDUCATION

Master of Science - Computer Science, Binghamton University 08/2023 – 08/2025 | Binghamton, NY
GPA: 3.8/4.0 | Relevant coursework: DSA, OS, **Distributed Systems, AI**, System Programming

Bachelor of Engineering - Electronics and Telecommunications, Pune University 08/2012 – 06/2016 | Pune, India

PROJECTS

AI-Powered Interview Evaluation Platform 01/2025 – 05/2025

- Built and deployed an AI-powered interview evaluation system using Java Spring Boot, RabbitMQ, and Mistral-7B, enabling automated question delivery, real-time response assessment, and personalized feedback.
- Integrated large language models (LLMs) with distributed messaging architecture to simulate technical and behavioral interviews, demonstrating advanced skills in real-time systems and AI workflow automation.

Real-Time Social Media Crawler for Political Discourse 01/2024 – 05/2024

- Developed real-time streaming pipelines in Python that ingested 1.8M 4chan posts and 0.95M Reddit comments in 14 days, enabling event-driven analysis without relying on standard crawling frameworks.
- Built custom CLI tools leveraging **Named Entity Recognition (NER)**, **sentiment analysis**, and **LDA topic modeling** to enable interactive querying and visualization of toxic discourse trends around major political events.